

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

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Program/Batch	B083-Networkwalks
Date	17 September 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.

All commands were run in Kali Linux, including both the footprinting tools and the Zenmap network scan. Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux (VirtualBox VM)	Operating system environment used for all reconnaissance and scanning activities
WHOIS	Find domain registration details (registrar, dates, name servers)
WhatWeb	Fingerprint web technologies (server, CMS, plugins, IP)
Nslookup	Resolve the domain name to its IP address using DNS
curl -i	Read the HTTP response headers of the website

Tool	Purpose
Wafw00f	Detect whether a Web Application Firewall protects the site
theHarvester	OSINT enumeration of subdomains, ASNs, IPs and emails (used in place of DNSRecon, which was not installed on this build)
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IP addresses and MAC addresses

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using WHOIS, WhatWeb, Nslookup, Curl and Wafw00f, and used theHarvester in place of DNSRecon, which was not available on this Kali build. Each tool was used to collect a different type of information about the target.

First, I used WHOIS to obtain publicly available domain registration information. The domain is registered with GoDaddy.com, LLC, was created on 6 November 2019 and is set to expire on 6 November 2027. The registrant and technical contact details are masked behind Domains By Proxy, LLC, and the domain resolves through name servers NS6135/NS6136.HOSTGATOR.COM (with NS29/NS30.DOMAINCONTROL.COM also listed on the registrar's record). All four client-side domain locks (delete, renew, transfer, update prohibited) are enabled.

I then used WhatWeb to identify technologies used by the website. The results identified WordPress 7.1 and WP Download Manager 3.3.58, along with Apache as the web server, jQuery 3.7.1, Bootstrap 7.1, Google Tag Manager, and IP address 192.232.216.135.

Using Nslookup, I resolved the domain name to its IP address via Google's public DNS resolver (8.8.8.8). The result confirmed 192.232.216.135.

I used curl -i to inspect the HTTP response headers. This confirmed the Apache server and WordPress-specific behaviour (X-Redirect-By: WordPress - Really Simple Security, X-nginx-cache: WordPress), and exposed the WordPress REST API endpoint /wp-json/ along with a rel=shortlink link header.

Next, I used Wafw00f to determine whether a Web Application Firewall was protecting the website. The result identified ModSecurity (SpiderLabs).

I attempted to use DNSRecon to enumerate DNS records, but the command was not installed on this Kali build. As a substitute, I ran theHarvester (-b all) against the domain. It identified 3 ASNs (AS13335, AS31898, AS46606 — AS13335 is associated with Cloudflare), 3 IP addresses (192.232.216.135 and 172.67.198.228, the latter consistent with Cloudflare-fronted infrastructure), 1 contact email (info@networkwalks.com), and 20 hosts, including administrative and mail-related subdomains such as cpanel, webdisk, webmail, mail, ftp and several autodiscover variants.

4.2 Network Scanning with Zenmap

For the second activity, I used Zenmap (the Nmap GUI) inside my Kali VM to run a Ping Scan against my local /24 subnet using the command nmap -sn 10.0.0.0/24.

The scan covered 256 IP addresses in 7.09 seconds and identified 3 live hosts:

- 10.0.0.1 — host up, MAC address 52:55:0A:00:00:01 (vendor unknown), most likely the network gateway
- 10.0.0.2 — host up
- 10.0.0.3 — host up, my own device

After completing the scan, I opened the Topology section in Zenmap, which plotted the three discovered hosts around localhost. Since a ping scan does not include port or traceroute data, the connections were shown as "no traceroute information" dashed lines and each host was marked with fewer than 3 open ports on the legend's colour scale. I reviewed the Legend to confirm the icon meanings before saving the topology graphic.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress 7.1 and WP Download Manager 3.3.58	Attackers may use exposed technology/version information to identify software requiring further security review	Medium
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Provides information about the network location of the web service	Low
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed the /wp-json/ REST API endpoint	May assist technology fingerprinting and further enumeration	Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	Low
5	Subdomain & infrastructure information exposed	theHarvester identified 20 hosts (cPanel, webmail, mail, ftp, autodiscover subdomains), 3 ASNs and a contact email	Subdomain and ASN information can help build a broader infrastructure profile and reveal administrative access points	Medium
6	Multiple live hosts visible on local network	Zenmap ping scan identified 3 live hosts on the 10.0.0.0/24 subnet, including a MAC address on the gateway host	Unknown or unauthorized devices may potentially be present on a network	Medium

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address, subdomain or MAC address does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information.** Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.
2. **Keep software updated.** WordPress and its plugins (including WP Download Manager) should be regularly updated and reviewed against current security advisories.
3. **Review HTTP headers.** HTTP response headers should be reviewed to determine whether unnecessary technical information, such as exposed API endpoints, is being disclosed.
4. **Audit exposed subdomains regularly.** Administrative and mail-related subdomains (cPanel, webmail, webdisk, autodiscover, etc.) should be reviewed and restricted to authorized users where possible.
5. **Properly configure and monitor the WAF.** Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.

6. **Perform regular internal network discovery.** Organizations should periodically scan their own networks to identify active devices.
7. **Investigate unknown devices.** Any unexpected device or MAC address discovered during network scanning should be investigated and verified.
8. **Maintain network documentation.** Network topology and device information should be documented and updated regularly.
9. **Verify tool availability before an assessment.** Confirm required tools (e.g. DNSRecon) are installed ahead of time, and keep a documented fallback tool for each enumeration step.
10. **Perform security testing with authorization.** Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used WHOIS, WhatWeb, Nslookup, Curl and Wafw00f against the networkwalks.com domain, and substituted theHarvester for DNSRecon when the latter was unavailable. I learned how WHOIS can reveal registration and privacy-protection details, WhatWeb can identify web technologies and versions, Nslookup can resolve domain names, Curl can inspect HTTP headers and exposed endpoints, Wafw00f can identify a WAF, and theHarvester can enumerate subdomains, ASNs and contact emails even without a dedicated DNS enumeration tool.

In the network scanning activity, I used Zenmap to run a ping scan across my local /24 subnet, identify three live hosts, capture a MAC address for the likely gateway, and generate a network topology graphic.

The exercises showed me that information gathering is an important part of cybersecurity, and that when a preferred tool is missing, an equivalent tool can often achieve the same reconnaissance goal. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

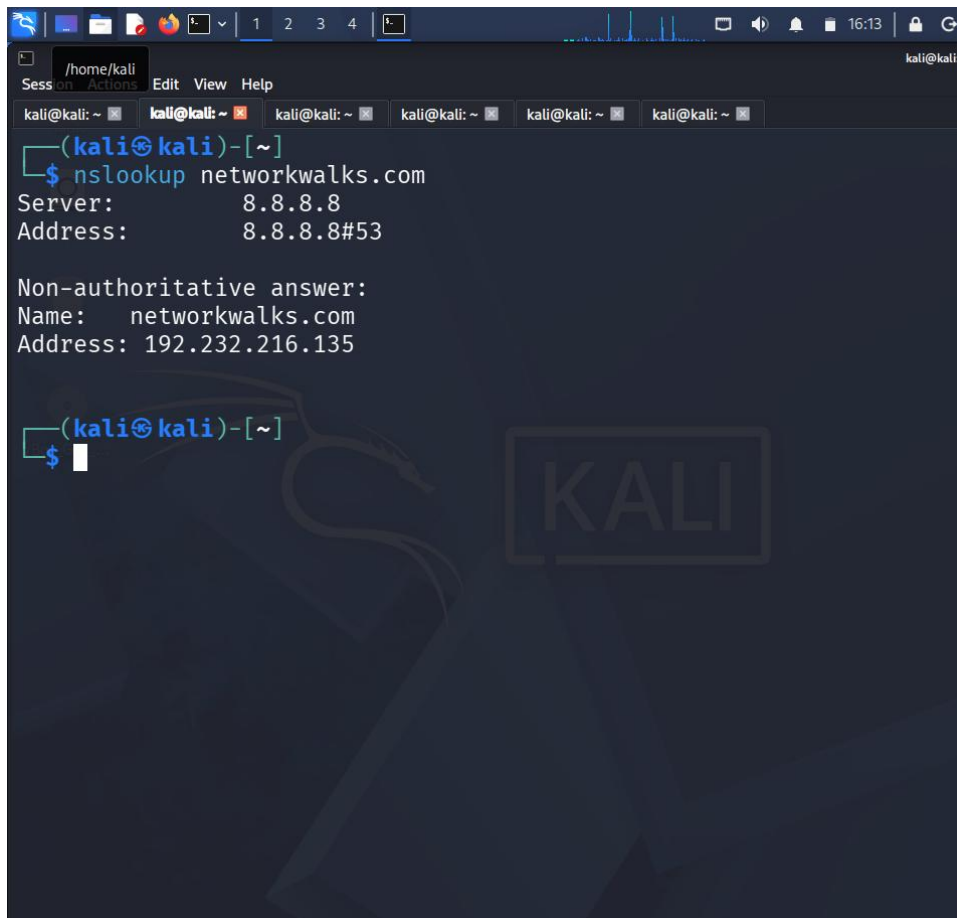
8. Evidences Collected

```
kali@kali: ~  
$ whois networkwalks.com  
Domain Name: NETWORKWALKS.COM  
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.godaddy.com  
Registrar URL: http://www.godaddy.com  
Updated Date: 2025-11-12T10:08:43Z  
Creation Date: 2019-11-06T22:51:46Z  
Registry Expiry Date: 2027-11-06T22:51:46Z  
Registrar: GoDaddy.com, LLC  
Registrar IANA ID: 146  
Registrar Abuse Contact Email: abuse@godaddy.com  
Registrar Abuse Contact Phone: 480-624-2505  
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited  
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited  
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited  
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited  
Name Server: NS6135.HOSTGATOR.COM  
Name Server: NS6136.HOSTGATOR.COM  
DNSSEC: unsigned  
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/whois-inaccuracy-complaint-form  
>>> Last update of whois database: 2026-09-13T20:21:06Z <<<  
  
For more information on Whois status codes, please visit https://icann.org/whois-status-codes  
  
NOTICE: The expiration date displayed in this record is the date the registrar's sponsorship of the domain name registration in the registry is currently set to expire. This date does not necessarily reflect the expiration date of the domain name registrant's agreement with the sponsoring registrar. Users may consult the sponsoring registrar's Whois database to view the registrar's reported date of expiration for this registration.
```

1. WHOIS lookup of networkwalks.com

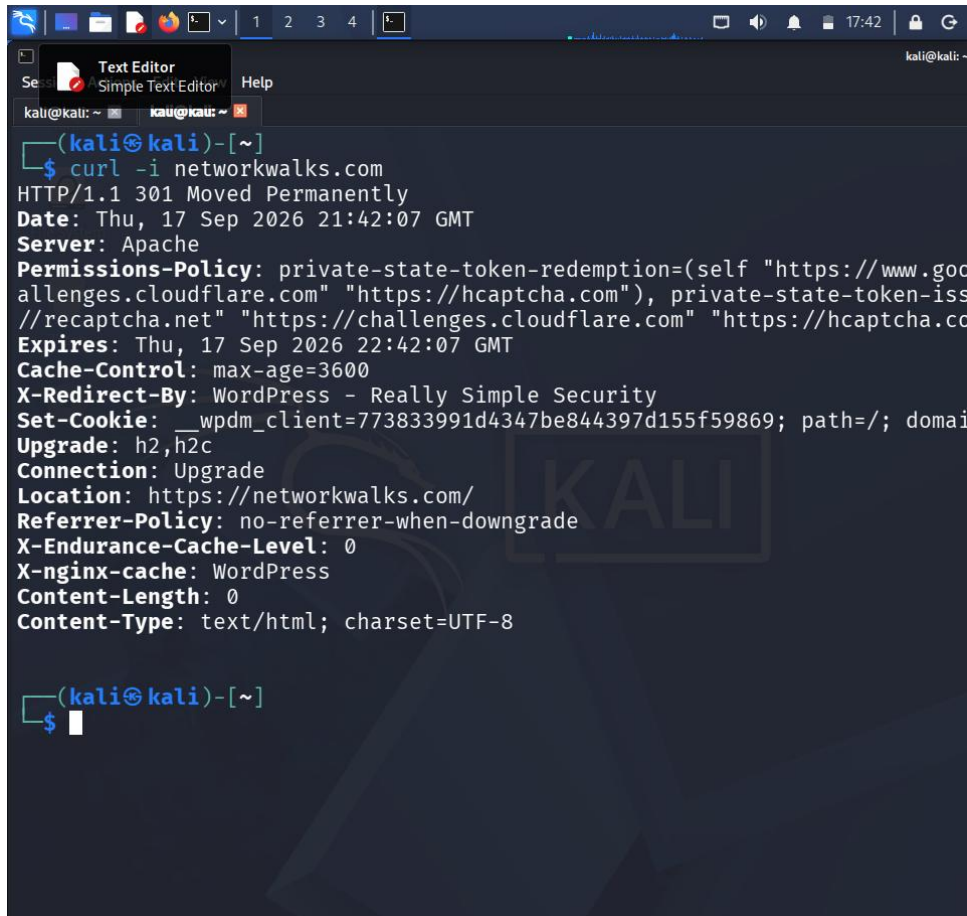
```
Session Actions Edit View Help
kali@kali: ~ kali@kali: ~ kali@kali: ~ kali@kali: ~ kali@kali: ~ kali@kali: ~
(kali@kali)-[~]
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpm_client],
[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders
level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1], Cookies[__wpm_client
ag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpm_client], IP[192.232.216.
3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPB&pidnVar2=50511&am
lationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[perm
cache], WordPress[7.1]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1], Country[UNITED STATE
Server[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1
84NR-IPB&pidnVar2=50511&pidnVar2=7&pidnVar2=12,application/json,appl
Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-en
```

2. WhatWeb technology fingerprint of networkwalks.com

A terminal window on a Kali Linux system. The window title is "/home/kali". The terminal shows the command `nslookup networkwalks.com` being executed. The output displays the server used (8.8.8.8) and the IP address for networkwalks.com (192.232.216.135). The prompt is `(kali㉿kali)-[~]`.

```
(kali㉿kali)-[~]  
$ nslookup networkwalks.com  
Server:      8.8.8.8  
Address:     8.8.8.8#53  
  
Non-authoritative answer:  
Name:   networkwalks.com  
Address: 192.232.216.135  
  
(kali㉿kali)-[~]  
$
```

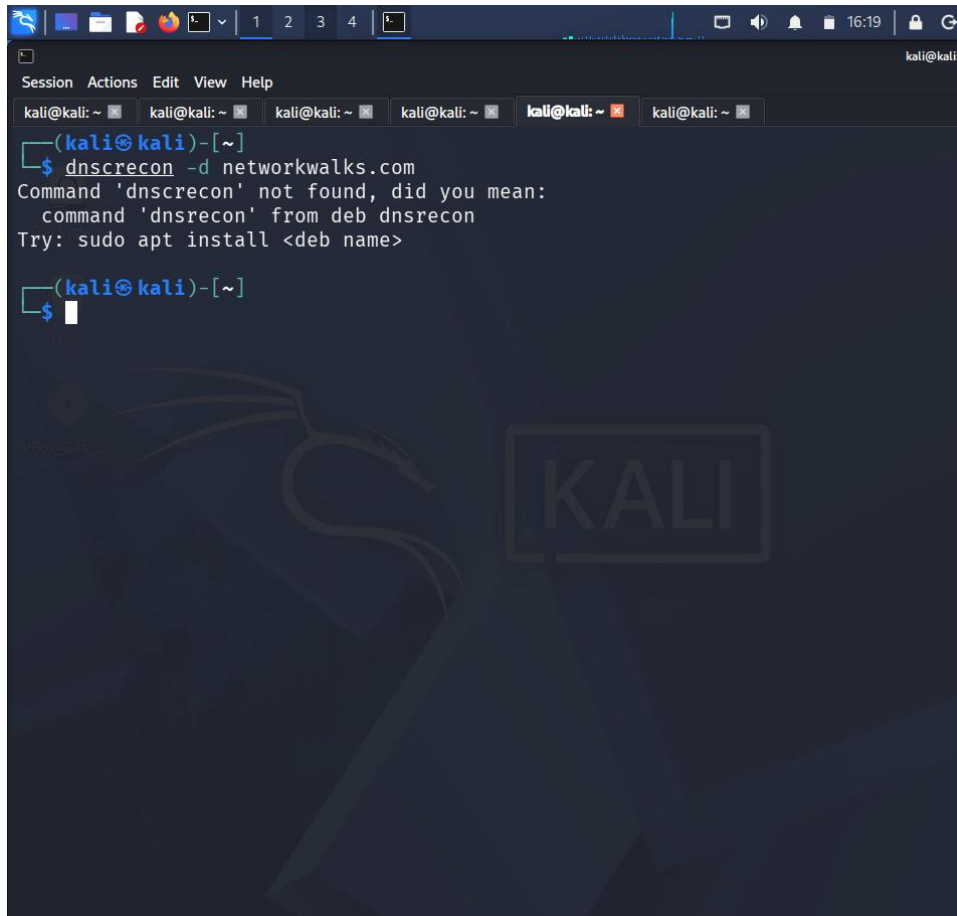
3. Nslookup resolving networkwalks.com



```
(kali㉿kali)-[~]
$ curl -i networkwalks.com
HTTP/1.1 301 Moved Permanently
Date: Thu, 17 Sep 2026 21:42:07 GMT
Server: Apache
Permissions-Policy: private-state-token-redemption=(self "https://www.google.com/recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com/recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
Expires: Thu, 17 Sep 2026 22:42:07 GMT
Cache-Control: max-age=3600
X-Redirect-By: WordPress - Really Simple Security
Set-Cookie: __wpdm_client=773833991d4347be844397d155f59869; path=/; domain=networkwalks.com
Upgrade: h2,h2c
Connection: Upgrade
Location: https://networkwalks.com/
Referrer-Policy: no-referrer-when-downgrade
X-Endurance-Cache-Level: 0
X-nginx-cache: WordPress
Content-Length: 0
Content-Type: text/html; charset=UTF-8

(kali㉿kali)-[~]
$
```

4. curl -i HTTP response headers



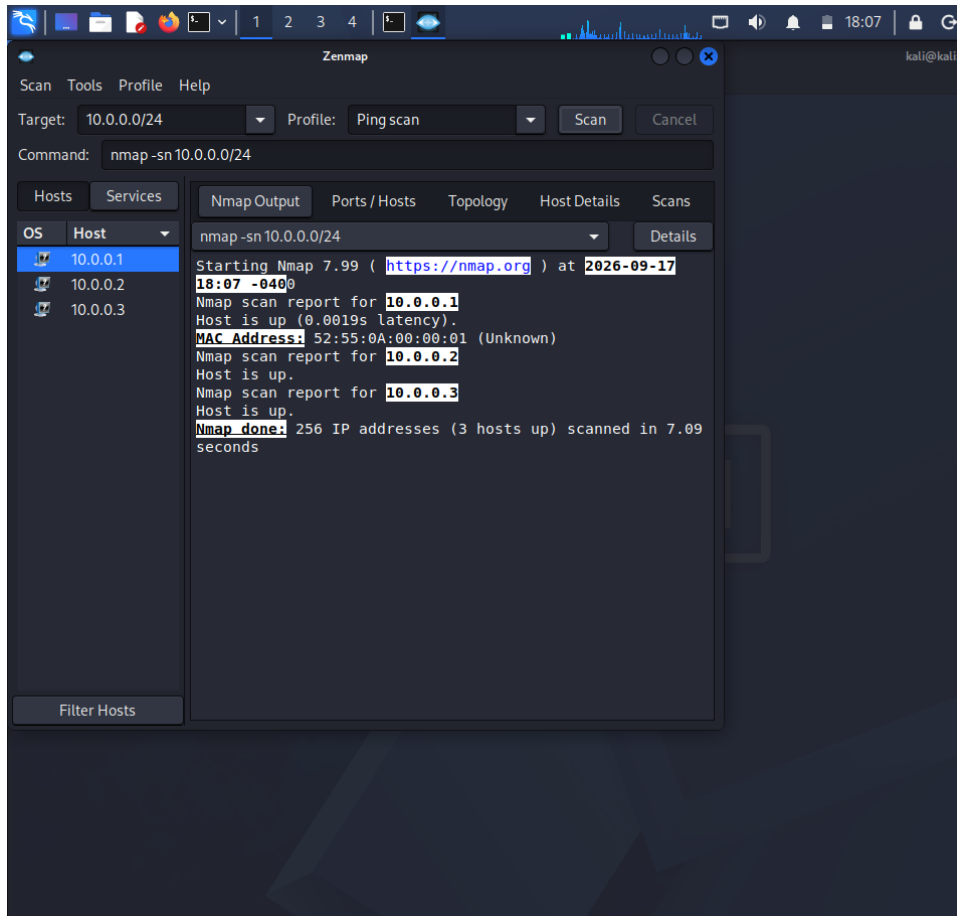
A terminal window on a Kali Linux system. The window has a dark blue background with a large, faint watermark of a dragon and the word "KALI". The terminal shows a prompt `(kali@kali)-[~]` followed by the command `$ dnscrecon -d networkwalks.com`. The output indicates that the command was not found and suggests installing the `dnsrecon` package. The prompt then returns to `(kali@kali)-[~]` with a new line starting with `$`.

```
(kali@kali)-[~]  
$ dnscrecon -d networkwalks.com  
Command 'dnscrecon' not found, did you mean:  
  command 'dnsrecon' from deb dnsrecon  
Try: sudo apt install <deb name>  
  
(kali@kali)-[~]  
$
```

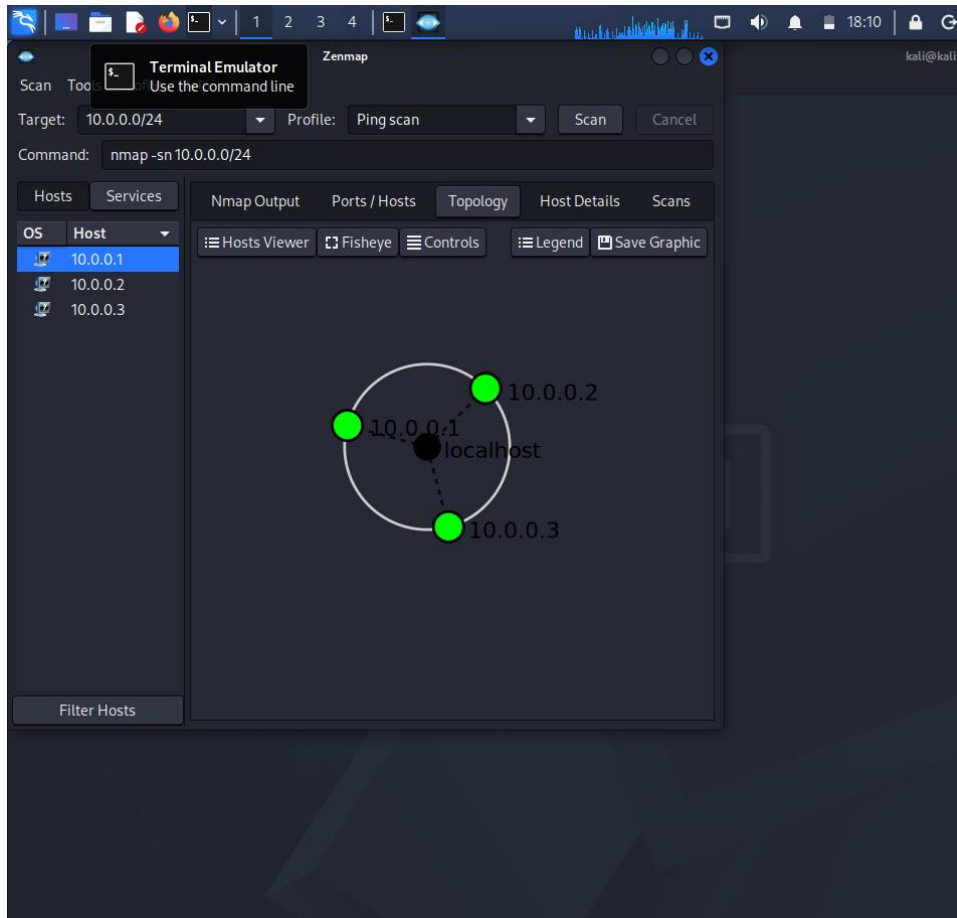
5. Wafw00f WAF detection result

```
kali@kali: ~  
$ wafw00f networkwalks.com  
  
? ??  
(( ))  
/,_/_/;  
\\_\\_\\  
  
~ WAFW00F : v2.4.2 ~  
~ Sniffing Web Application Firewalls since 2009 ~  
  
[*] Checking https://networkwalks.com  
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.  
[~] Number of requests: 2  
  
kali@kali: ~  
$
```

6. DNSRecon unavailable on this build (command not found)



8. Zenmap/Nmap ping scan output (10.0.0.0/24)



9. Zenmap network topology of discovered hosts